**1** A student investigates the vertical oscillations of a solid cylinder which floats in cooking oil. Fig. 1.1 shows a cylinder of radius *r*.

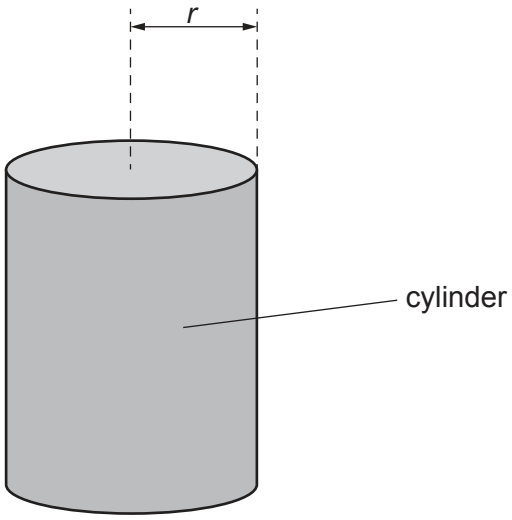


**Fig. 1.1**

The student places the cylinder of mass *m* in the oil. The cylinder is displaced vertically from its equilibrium position and released so that it oscillates. The period *T* of the oscillations is determined.

A number of cylinders of different mass are available.

It is suggested that the relationship between *T* and *m* is

$$T = 2\sqrt{\frac{\pi m}{\sigma K r^2}}$$

where $\sigma$ is the density of the oil and *K* is a constant.

Design a laboratory experiment to test the relationship between *T* and *m*.
Explain how your results could be used to determine a value for *K*.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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